

Student Pass/Fail Prediction

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Problem Statement

- Many students fail due to lack of early identification
- Manual tracking is difficult and time-consuming
- Need a system to **predict student performance in advance**
- **Why it is important:**
- Helps teachers identify weak students early
- Enables timely intervention and support
- Improves overall academic performance

Project Explanation

- Built a machine learning model to predict:
 - **Pass (1) or Fail (0)**
- Used student-related data for prediction
- **Dataset includes:**
 - Attendance
 - Study hours
 - Previous marks
 - Assignments
- **Workflow:**
Data Collection → Data Preprocessing → Model Training → Prediction → Results

Model / Algorithms Used

- Logistic Regression
- Decision Tree

Why these models:

- Suitable for classification problems
- Easy to understand and implement
- Provide good accuracy on structured data

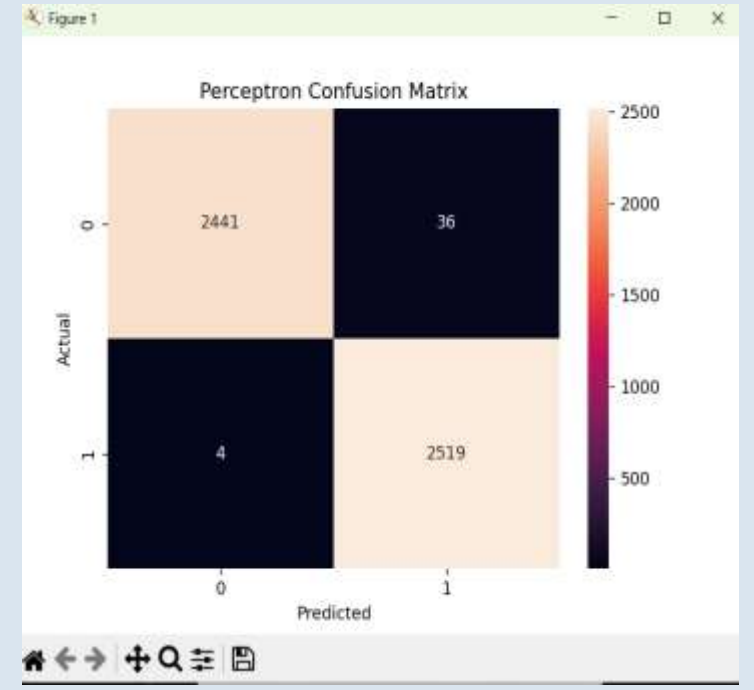
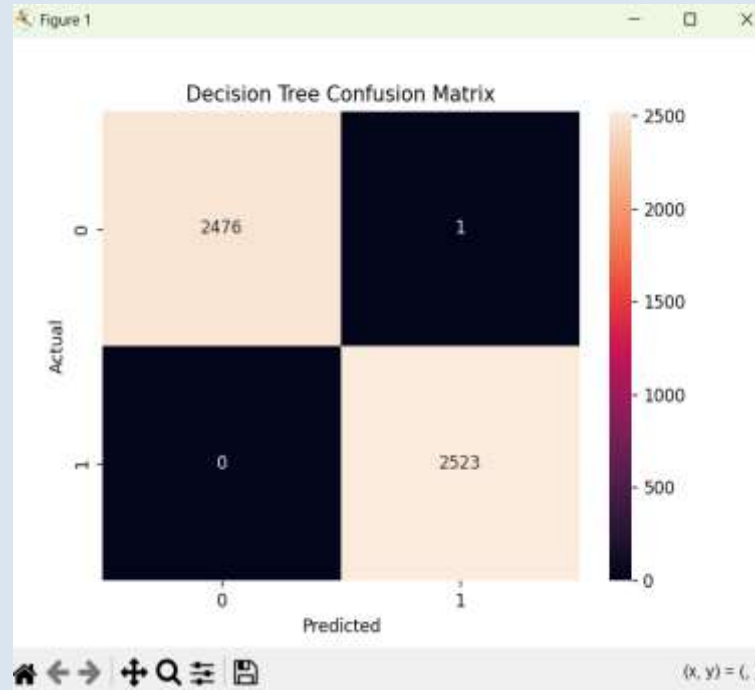
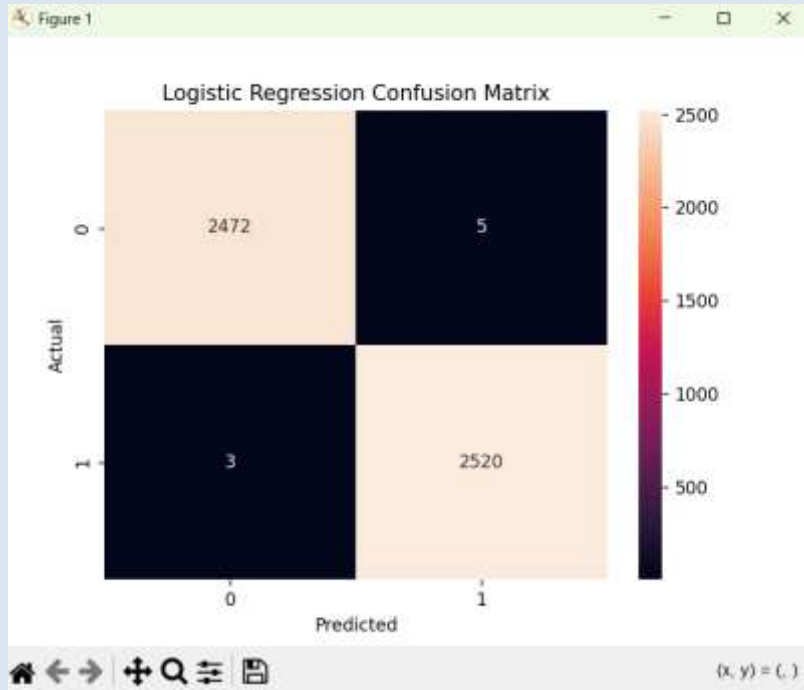
Performance Metrics

- **Accuracy** – Overall correctness of model
- **Precision** – Correct positive predictions
- **Recall** – Ability to find all positive cases
- **F1-Score** – Balance between precision and recall

Confusion Matrix:

- True Positive, True Negative
- False Positive, False Negative

Figures/Plots



Outcome & Conclusion

- Model successfully predicts student performance
- Achieved good accuracy (mention your value)
- Helps in identifying at-risk students early

What worked:

- Proper data preprocessing improved accuracy

What didn't:

- Limited dataset size affects performance

Future improvements

- Use larger and real-time datasets
- Apply advanced models (Random Forest, Neural Networks)
- Build a web-based prediction system

https://github.com/harshitha-reddy-06/Machine-learning/blob/main/student_prediction.py



Thank you